

Challenge: Building a Real Estate Pricing Model API

Description:

Your challenge is to design and implement a robust real estate pricing model that accurately predicts property prices. The model should address real-world complexities, such as data quality, feature interactions, and performance, and be accessible through an API for seamless integration into various applications.

Problem Focus:

Real estate pricing is inherently complex, influenced by numerous factors such as location, property features, market trends, and economic conditions. The challenge lies in:

1. Data Quality Issues:

- Real estate data can be incomplete, inconsistent, or outdated, requiring careful preprocessing and cleaning.

2. Model Complexity:

- Capturing nuanced relationships between property features and prices.
- Accounting for external economic indicators like inflation or interest rates.

3. Operational Usability:

- Exposing the model predictions through an API that is intuitive, reliable, and scalable.
 - Enabling easy retraining with new or updated data.
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Requirements:

1. Problem Understanding and Data Handling

- Analyze the core factors driving house prices.
- Address data inconsistencies, missing values, and outliers.

2. Predictive Model Development

- Explore and compare algorithms

- Select and fine-tune the model based on performance metrics such as RMSE, R^2 , or MAE.
- Ensure the model generalizes well through cross-validation.

3. API Integration

- Build a RESTful API using Python frameworks like Flask or FastAPI.
- The API should accept property features as input and return a predicted price.
- Implement robust error handling to manage invalid inputs or system issues.

4. Operationalization and Documentation

- Provide step-by-step instructions for setting up the API and dependencies.
 - Document procedures for retraining the model with updated data.
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What You'll Need to Deliver:

- A trained and optimized pricing model.
 - Documentation for developers and users.
 - (bonus) A working API that exposes the model predictions.
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Datasets (Optional):

1. **properties.csv**: Property attributes such as location, size, and features.
2. **transactions.csv**: Historical sale prices and transaction details.
3. **economics.csv**: Broader economic trends, including interest rates and inflation.

Note: While these datasets are provided, you are encouraged to focus on solving the real estate pricing problem effectively, leveraging any data sources that add value.